Kyle Aaron Thompson

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Education

Auburn University — Auburn, Alabama

Ph.D. Mechanical Engineering, Minor in Aerospace Engineering — *Expected 2028*

M.S. Mechanical Engineering — *August 2025*

B.S. Mechanical Engineering, Magna Cum Laude — *August 2020*

- **Graduate GPA:** 4.0/4.0
- **Graduate Coursework:** Optimal Control of Aerospace Vehicles, Advanced Trajectory Optimization, Nonlinear Systems and Control, Adaptive Control, Advanced Discrete Control, Fundamentals of Navigation and Guidance, Fundamentals of GPS, Neural Networks, Missile Aerodynamics

Experience

Graduate Research Assistant – Auburn University — August 2020 – Present

GPS and Vehicle Dynamics Laboratory (GAVLAB) — *Auburn, Alabama*

- Conduct sponsored research in guidance, navigation, and control (GNC) of autonomous ground vehicles, focusing on guidance and control, with additional work on aerial platforms.
- Developed longitudinal and lateral guidance and control for Class 8 trucks, sedans, and racing platforms.
- Serve as controls, guidance, and software lead for Autonomous Tiger Racing (ATR), Auburn's Indy Autonomous Challenge (IAC) team.
- Designed raceline trajectory optimizers formulated as nonlinear programs.
- Built real-time optimal feedback controllers as quadratic programs solved online with OSQP for reliable timing.
- Published multiple academic papers as first author and co-authored additional conference and journal papers.

United States Marine Corps Reserves — July 2014 – Present

Gunnery Sergeant

- Oversee safe operation of M142 HIMARS rocket artillery systems for Battery K, 2nd Battalion, 14th Marines.
- Developed and taught curriculum for the professional certification of HIMARS operators.
- Deployed to Afghanistan with Task Force Southwest 19.1 and 19.2 (2019) and to Okinawa (2022–2023).

Teledyne Brown Engineering, Inc. — April 2015 – July 2015; June 2016 – August 2016

Classified Lab Administrative Assistant — *Huntsville, Alabama*

- Opened and secured the classified lab after hours and enforced government and company security policies.
- Performed basic system administration for classified lab users and supported software documentation.

Selected Publications

- **K. Thompson**, "Intelligent Emergency Stop System for Autonomous Ground Vehicles Using Minimum Jerk Trajectories and Path Tracking Linear Quadratic Regulator," *M.S. thesis*, Auburn University, Auburn, AL, Aug. 2025.
- **K. Thompson**, B. Jones, S. Martin, and D. Bevly, "GPS-Independent Autonomous Vehicle Convoying with UWB Ranging and Vehicle Models," *NDIA Ground Vehicle Systems Engineering and Technology Symposium (GVSETS)*, Novi, MI, Aug. 2022.

Skills

Guidance and Control: Classical control, LQR, MPC, trajectory optimization, path tracking, Kalman filtering
Languages: C++, Python, MATLAB; also C# (Windows), Swift (Apple), and Kotlin (Android)
Tools and Libraries: ROS/ROS 2, Qt, CasADi, IPOPT, OSQP, Git, LaTeX, and SolidWorks (3D CAD)
Computer and Software: Linux, Windows, macOS, Visual Studio, Xcode
Embedded Systems: Autonomy software on single-board computers and microcontrollers